

Comparative Analysis of Sequential Models for Mobile Network Traffic Forecasting

Formative 1 Assignment

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Abstract

Mobile operators need short-term traffic forecasts to manage network resources and congestion. This study looks at Internet traffic in Milan, where demand changes strongly by time of day and location.

The study uses the Telecom Italia Milan CDR dataset, covering 10,000 grid cells at 10-minute intervals over about two months. The data shows a strong 24-hour pattern and large differences between busy and quiet areas.

Three models were compared: SARIMA with Fourier-term seasonality, a gradient-boosted tree model using lag and calendar features, and an LSTM neural network. They were tuned for one week and tested on 16-22 December 2013.

The results show that the gradient-boosted model was strongest for the busiest area, while SARIMA was more consistent and performed best in the two lower-traffic areas. The LSTM performed poorly on the busiest area and required much more computing time. The main takeaway is that model performance depends on the traffic pattern of the area.

1. Introduction

Mobile network operators use traffic forecasts to plan resources, manage congestion, and prepare network capacity. Traffic can change a lot depending on the time of day, day of the week, and location.

This study uses the Telecom Italia Big Data Challenge dataset for Milan. It records SMS, call, and Internet activity across a 10,000-cell grid every 10 minutes.

The research question is: “How do different sequential models compare for one-step-ahead mobile network traffic forecasting, and how does their performance vary across geographical areas with different traffic characteristics?”

The aim is not simply to find one model with the smallest error. Instead, the study describes the data, compares three different forecasting approaches, and explains where and why each model works well or poorly.

2. Related Work

Previous research generally uses three main types of models for cellular traffic forecasting: statistical models, tree-based machine learning, and recurrent deep learning.

Statistical models

SARIMA is a common forecasting method because it can represent trends, seasonality, and short-term relationships with relatively few parameters. For high-frequency data, Fourier terms can represent the 144-step daily cycle more efficiently than using a very large seasonal model.

Deep learning

LSTM networks are often used for traffic forecasting and can work well when there is a large amount of detailed training data. Research on the same Milan dataset has reported strong LSTM results. Therefore, the weak LSTM results in this study are more likely related to the particular forecasting setup than to LSTM being unsuitable in general.

Tree-based / gradient boosting

Gradient-boosted trees can work well when a time series is converted into useful features such as recent traffic values, previous-day traffic, hour of day, and day of week. This study uses this approach to test whether a flexible machine-learning model can capture the same patterns.

Model selection

The exploratory analysis showed a strong 24-hour cycle, stationarity after accounting for seasonality, and large differences in traffic between areas. Based on these findings, SARIMA, gradient boosting, and LSTM were selected as three clearly different forecasting approaches.

3. Dataset and Data Preparation

The dataset contains 62 daily files covering 1 November 2013 to 1 January 2014. It contains 10,000 geographical grid cells with measurements every 10 minutes. The raw data includes SMS, calls, and Internet activity, with separate rows for different country codes.

3.1 Memory-efficient loading

The dataset is very large, so a memory-efficient loading process was used. Only the square ID, time interval, and Internet activity columns were kept. Smaller data types were used, and the files were processed in chunks. Internet activity was also aggregated across country codes before the data was combined.

The optimized process produced 89,245,318 aggregated rows using about 1,427.9 MB of memory. A full naive loading test was not performed because it was expected to require far more memory. The trade-off is that SMS and call columns were not kept in the working dataset and would need to be loaded again if required.

3.2 Continuity handling

All 62 expected daily files were present. A detailed check also confirmed all 8,928 expected 10-minute time steps, with zero missing intervals. This is important because forecasting models depend on a regular time sequence.

4. Exploratory Analysis

The three areas with the highest total traffic were squares 5161, 5059, and 5259. The assignment also specified areas 4159 and 4556. These five areas were used to understand different traffic levels, while 5161, 4159, and 4556 were carried forward to the forecasting experiments.

4.1 Distribution of total traffic

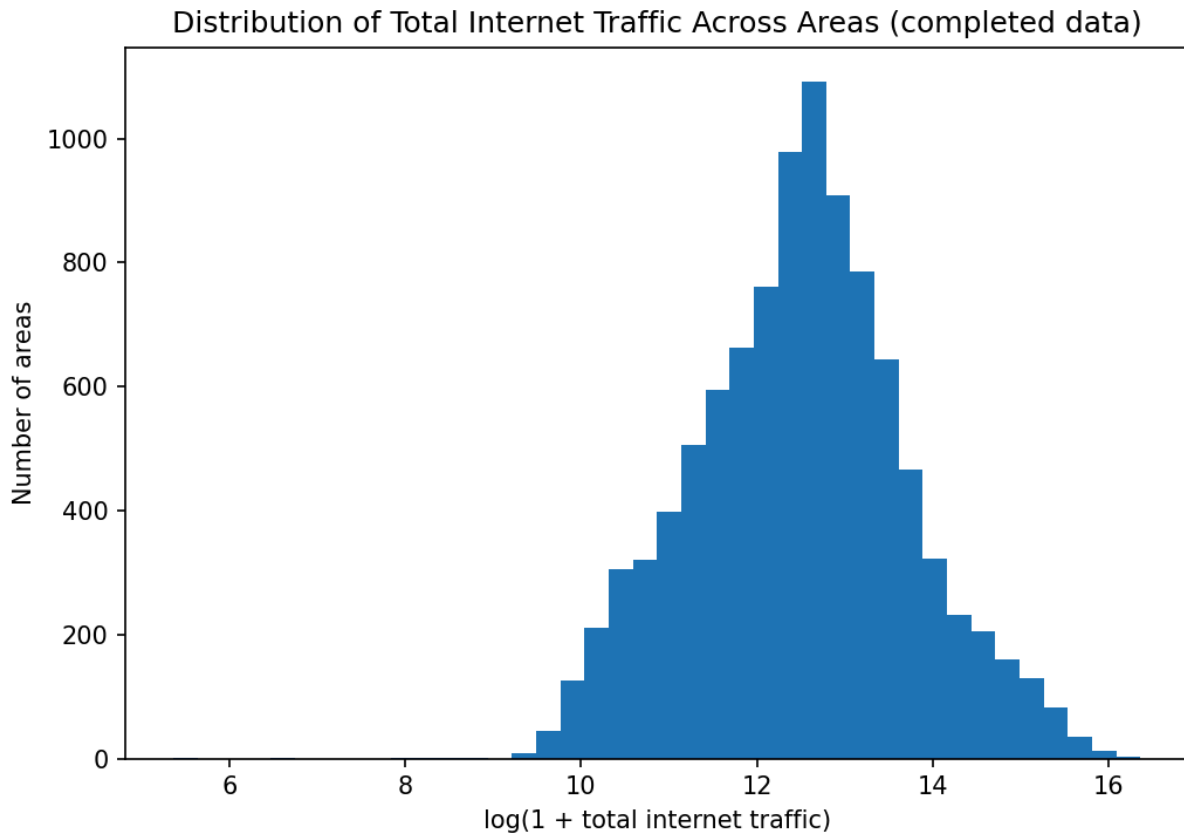


Figure 1. Distribution of total Internet traffic (log-scaled) across all 10,000 grid areas.

Most areas had moderate traffic, while a small number had much higher traffic. This difference between areas is why the models were tested on both busy and quieter locations.

4.2 Temporal dynamics - first two weeks

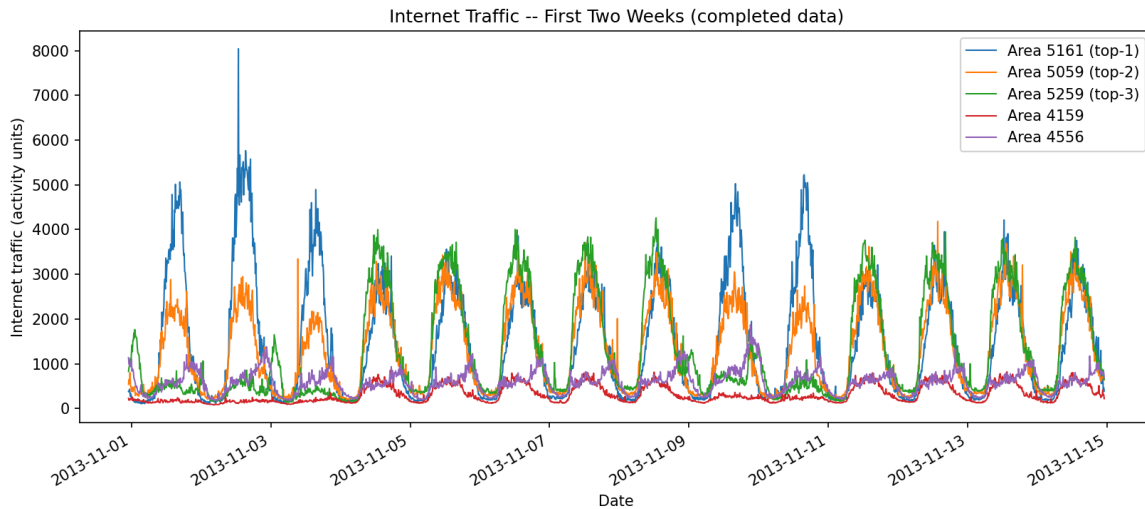


Figure 2. Internet traffic during the first two weeks for the three highest-traffic areas plus the two assignment-specified areas (4159, 4556).

All five areas show a strong daily pattern: traffic is lower at night and higher during the day. The cycle repeats about every 144 ten-minute steps, which equals 24 hours.

Area 5161 was especially volatile, with sharp peaks reaching about 8,000 units. Areas 5059 and 5259 had smoother peaks around 3,500-4,000. Areas 4159 and 4556 had much lower traffic but followed the same daily pattern.

4.3 Stationarity and autocorrelation

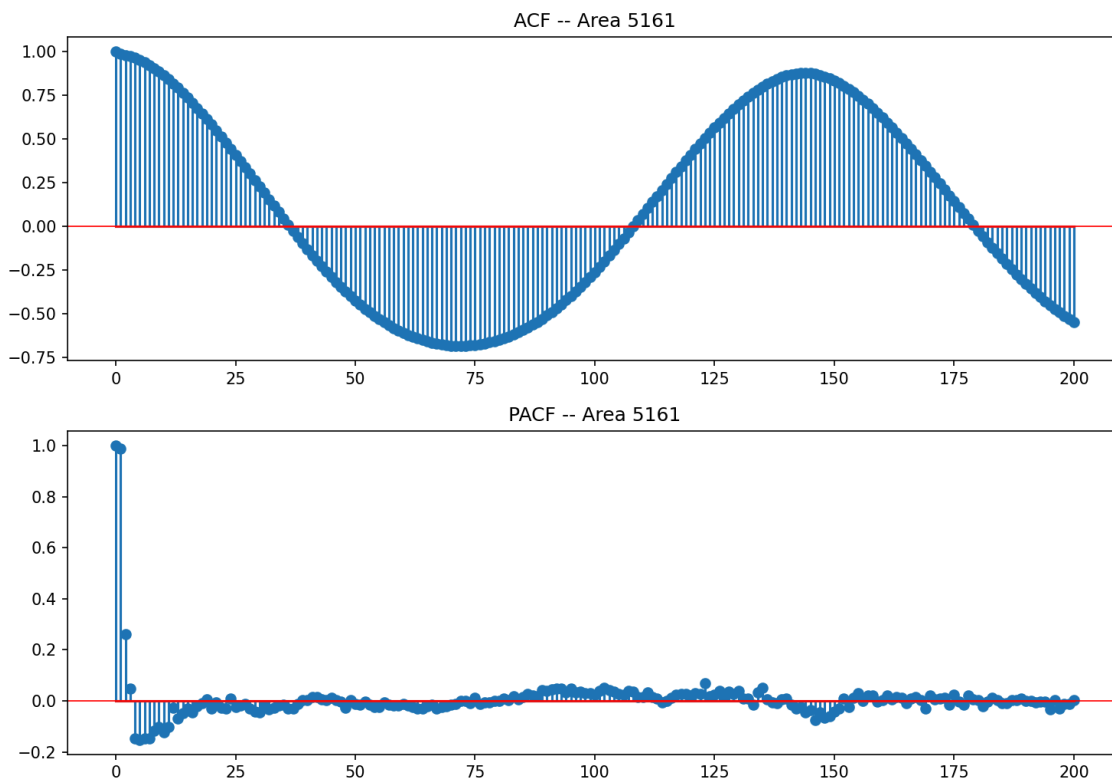


Figure 3. ACF and PACF for the highest-traffic area (5161), 200 lags (~33 hours).

The Augmented Dickey-Fuller test indicated that the highest-traffic series was stationary once its strong seasonal structure was considered. The ACF showed a clear repeat around lag 144, confirming the 24-hour cycle.

These results supported treating the series as stationary and representing daily seasonality explicitly instead of using a very large native seasonal order.

4.4 Seasonal decomposition

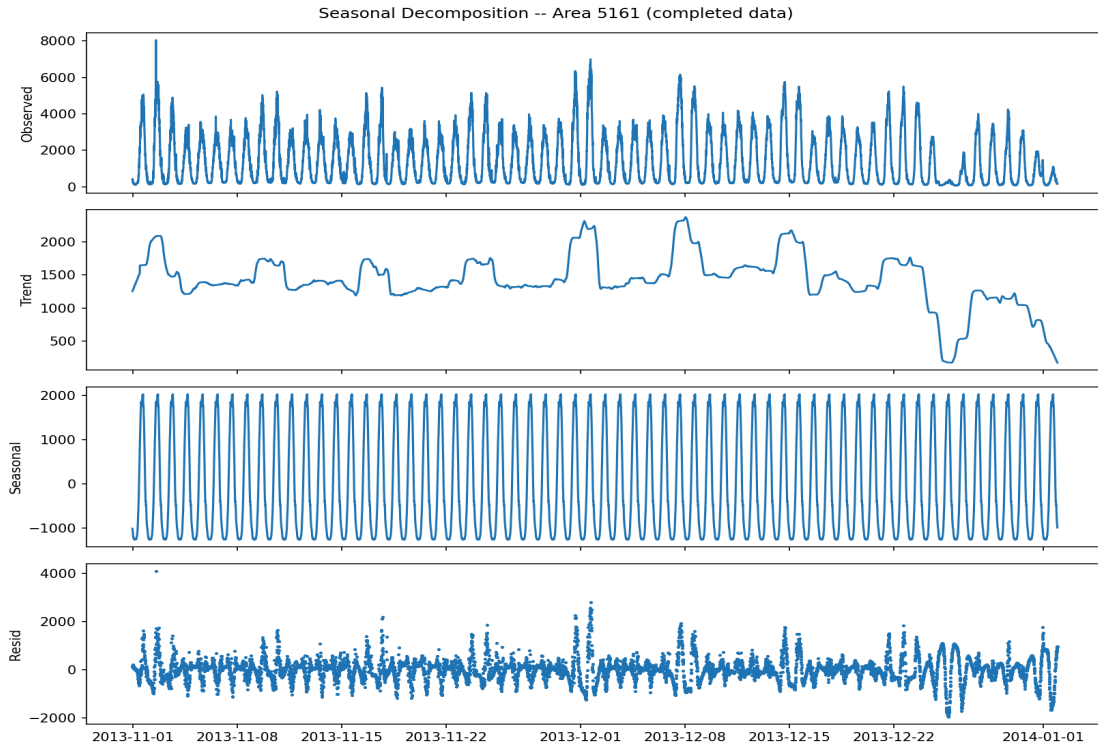


Figure 4. Additive seasonal decomposition of the highest-traffic area over the full observation period.

The decomposition shows a stable daily seasonal pattern, with an amplitude of roughly $\pm 2,000$ units, together with a slower-changing trend. Traffic drops toward the end of December, which matches the holiday period. The residuals also show some unusually large values.

4.5 Anomaly detection

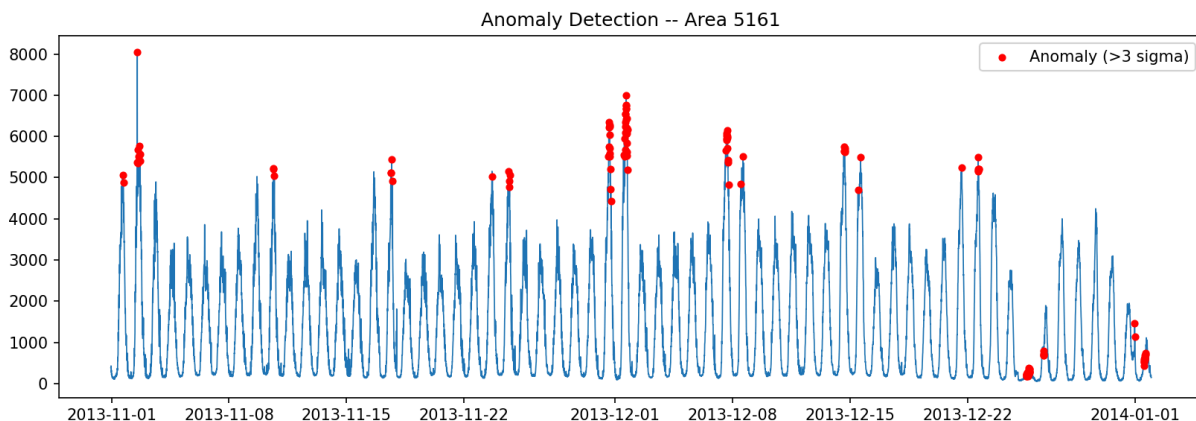


Figure 5. Points exceeding 3 standard deviations of the decomposition residual, highest-traffic area.

Using a 3-standard-deviation rule, 138 unusual 10-minute intervals were identified. These were concentrated on a small number of days. Such unusual peaks can be difficult for forecasting models because they are not common in the training data.

5. Methodology

The study compared SARIMA, gradient-boosted trees, and LSTM. All models were trained using a log1p transformation of traffic values. Predictions were converted back to the original scale and clipped at zero so that negative traffic values were not produced.

5.1 SARIMA (with Fourier-term seasonality)

A normal SARIMA seasonal period of 144 would be computationally expensive. Instead, Fourier terms were used to represent the daily pattern. The SARIMA order and number of Fourier harmonics were selected through grid search.

5.2 Gradient-boosted trees

LightGBM was originally intended for this part of the study. However, its OpenMP dependency could not be loaded on the development machine, so a comparable scikit-learn gradient-boosting regressor was used instead. For accuracy, these results are called “GBM-fallback,” not LightGBM results.

The model used 18 recent lags, one-day and two-day lags, and cyclical hour-of-day and day-of-week features. Multi-step predictions were made recursively, meaning each prediction was used as an input for the next prediction.

5.3 LSTM

The LSTM used a sliding window of previous traffic values and predicted the next value. Multi-step forecasts were also produced recursively. Predictions were clipped before being added back to the input sequence.

5.4 Hyperparameter tuning

All models were tuned using the validation week of 9-15 December 2013. The final evaluation week, 16-22 December, was not used during model selection.

The selected configurations were:

- SARIMA: order (2,1,2) with 3 daily Fourier harmonics.
- GBM-fallback: 31 leaves, learning rate 0.05, and 300 estimators.
- LSTM: 288-step lookback, 32 units, and 20 epochs.

The validation results were:

Model/configuration	MAE	RMSE	MAPE	sMAPE
SARIMA (2,1,2), K=3	550.39	854.48	28.28%	34.32%
GBM-fallback (31, 0.05, 300)	251.26	384.47	17.01%	16.21%

LSTM (288, 32, 20)	1018.49	1504.15	95.15%	68.88%
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5.5 Evaluation protocol

The models were retrained using all data before the evaluation period and tested on three areas: 5161, 4159, and 4556. Performance was measured using MAE, RMSE, MAPE, and sMAPE. sMAPE was included because MAPE can become unstable when traffic is close to zero.

6. Results and Discussion

The evaluation was performed using Python 3.9.6 on macOS with 16 logical CPUs and 68.7 GB of RAM. No GPU was used.

6.1 Highest-traffic area (square 5161)

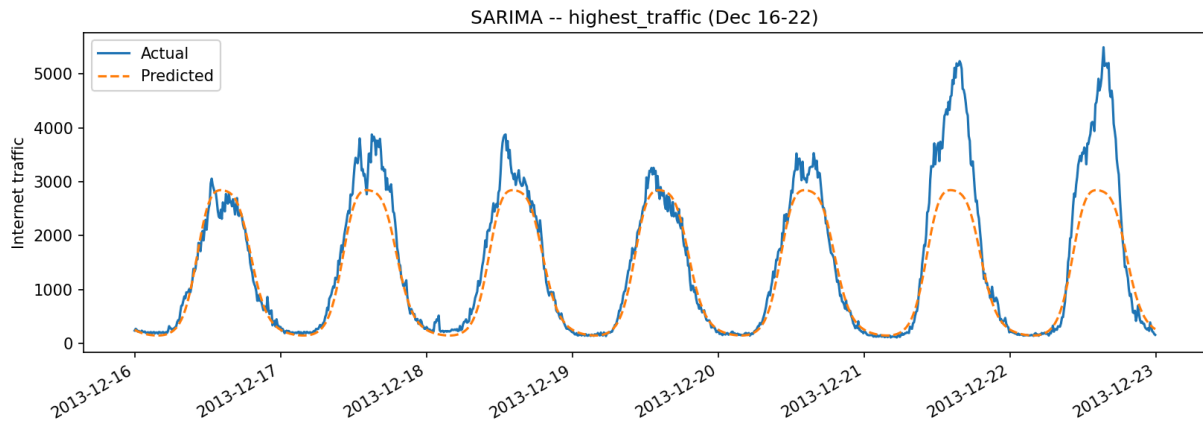


Figure 6a. SARIMA forecast vs. actual, highest-traffic area, Dec 16-22.

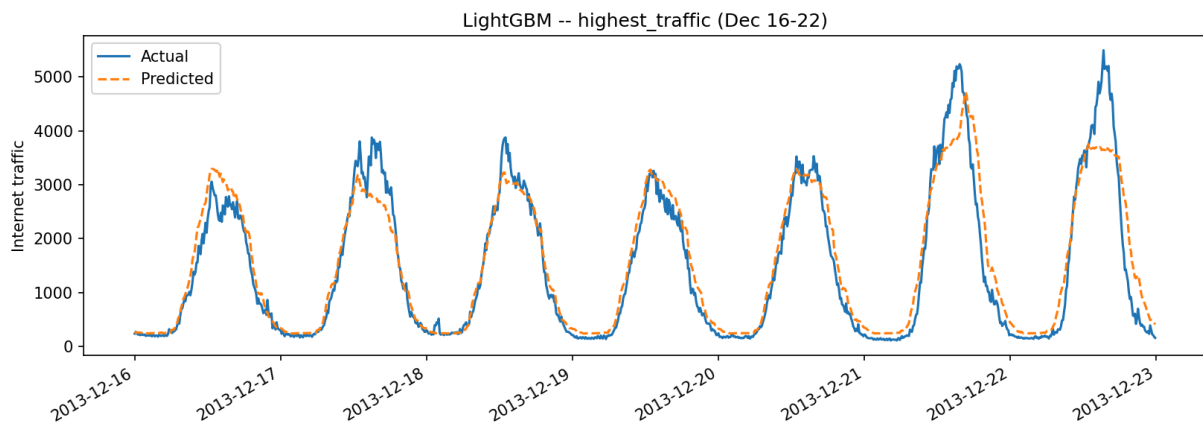


Figure 6b. GBM-fallback forecast vs. actual, highest-traffic area, Dec 16-22.

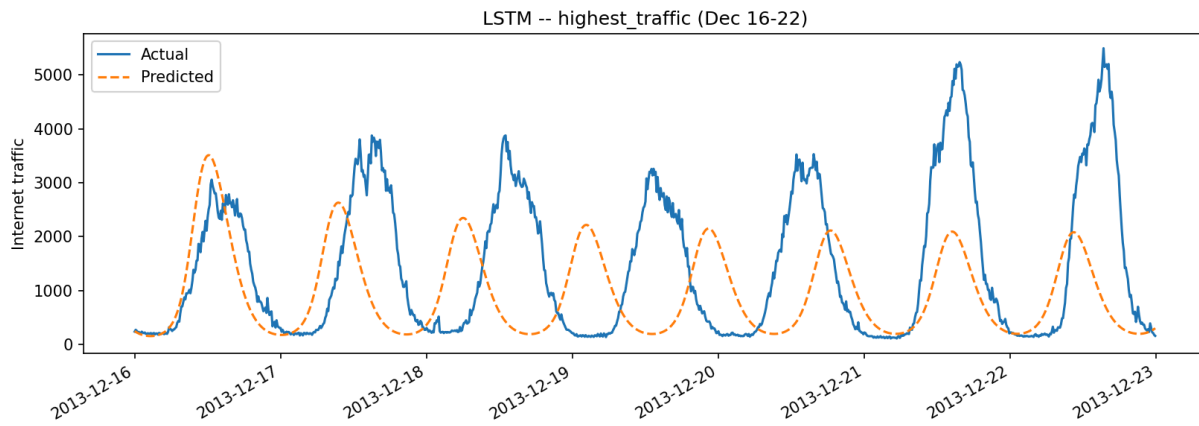


Figure 6c. LSTM forecast vs. actual, highest-traffic area, Dec 16-22.

Model	MAE	RMSE	MAPE	sMAPE
SARIMA	294.28	540.93	19.77%	20.78%
GBM-fallback	259.49	385.98	32.31%	25.46%
LSTM	1161.15	1535.96	153.99%	91.29%

In this busy area, GBM-fallback had the lowest MAE and RMSE and followed the daily traffic shape more closely. SARIMA was steady but compressed the size of the peaks. LSTM performed much worse and gradually produced a low-amplitude pattern instead of following the large peaks.

6.2 Square 4159 (low-traffic area)

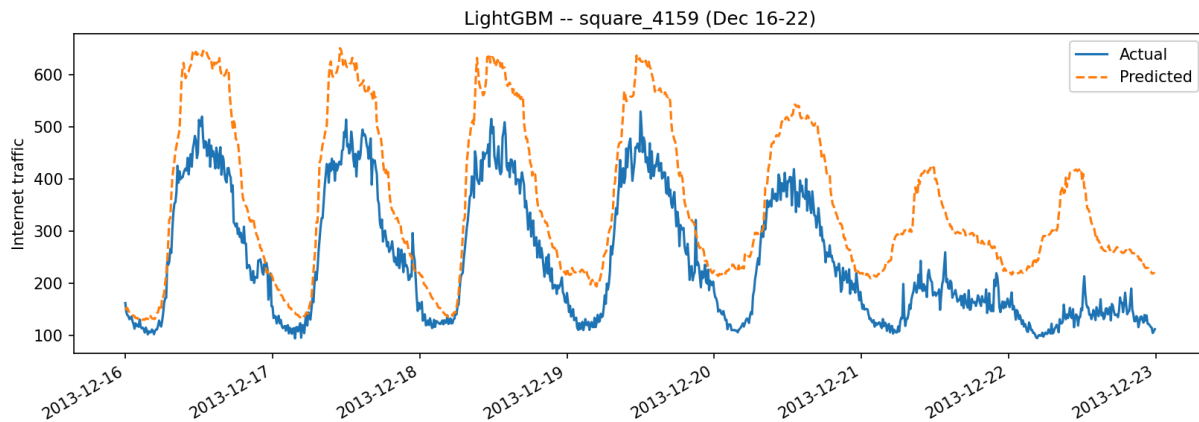


Figure 7a. SARIMA forecast vs. actual, square 4159, Dec 16-22.

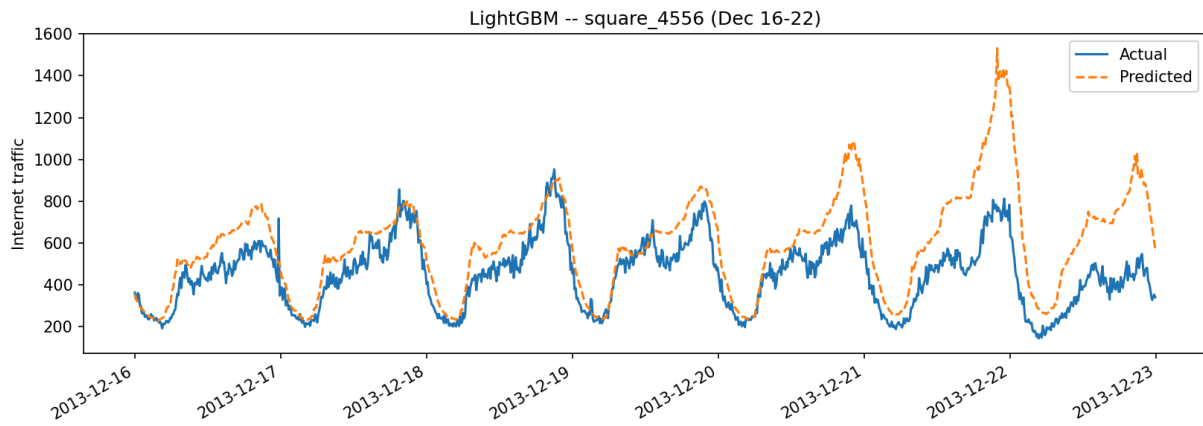


Figure 7b. GBM-fallback forecast vs. actual, square 4159, Dec 16-22.

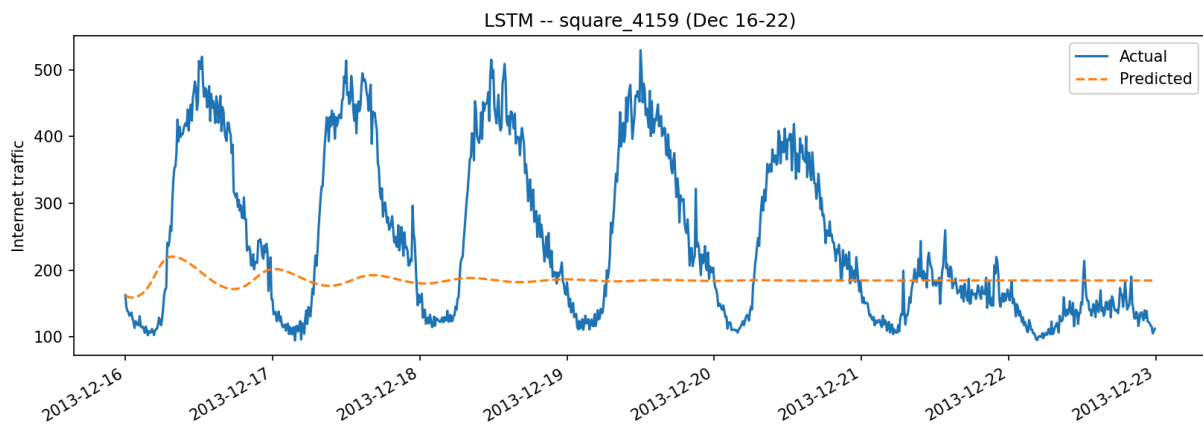


Figure 7c. LSTM forecast vs. actual, square 4159, Dec 16-22.

SARIMA had the lowest MAE and RMSE here: 62.71 and 82.63. GBM-fallback had MAE 111.64 and RMSE 126.31, while LSTM had MAE 101.48 and RMSE 134.28. This shows that model performance changed when the traffic level was lower.

6.3 Square 4556 (low-traffic area)

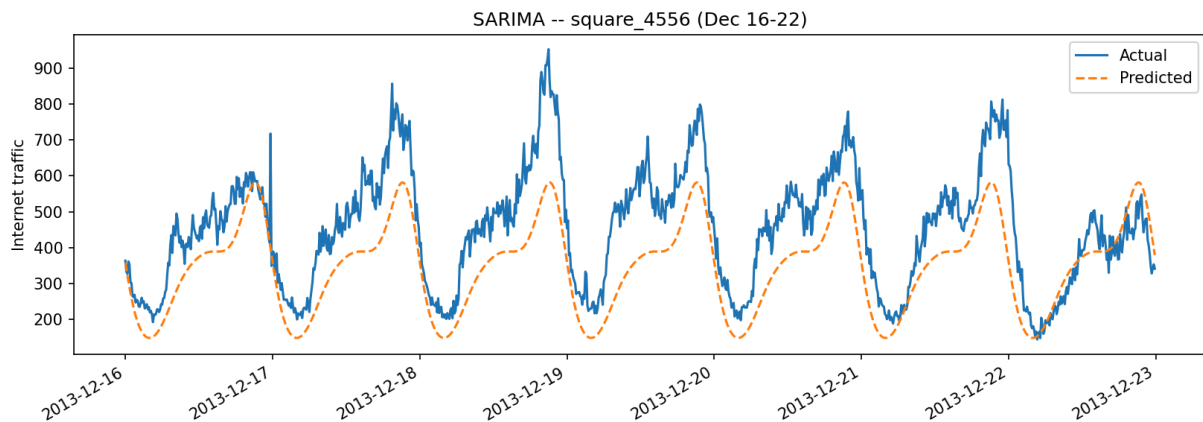


Figure 8a. SARIMA forecast vs. actual, square 4556, Dec 16-22.

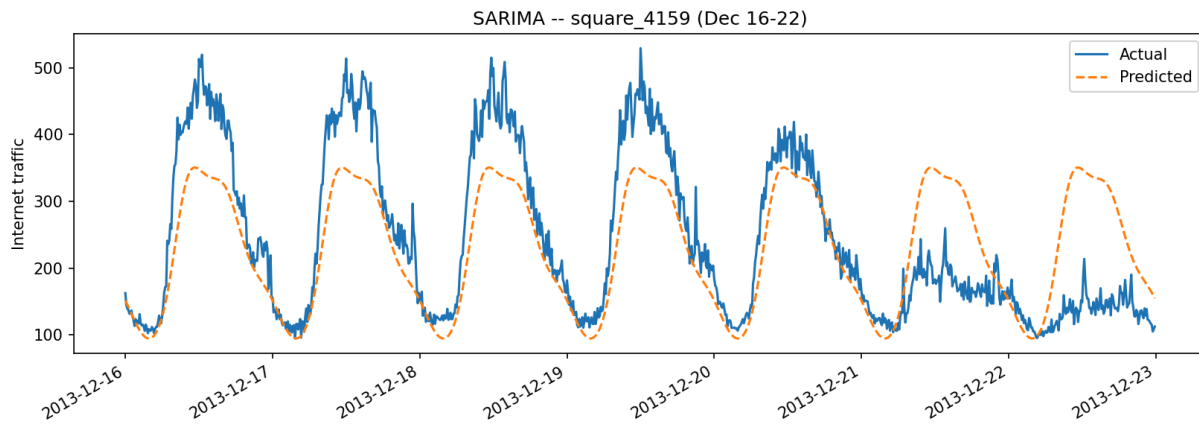


Figure 8b. GBM-fallback forecast vs. actual, square 4556, Dec 16-22.

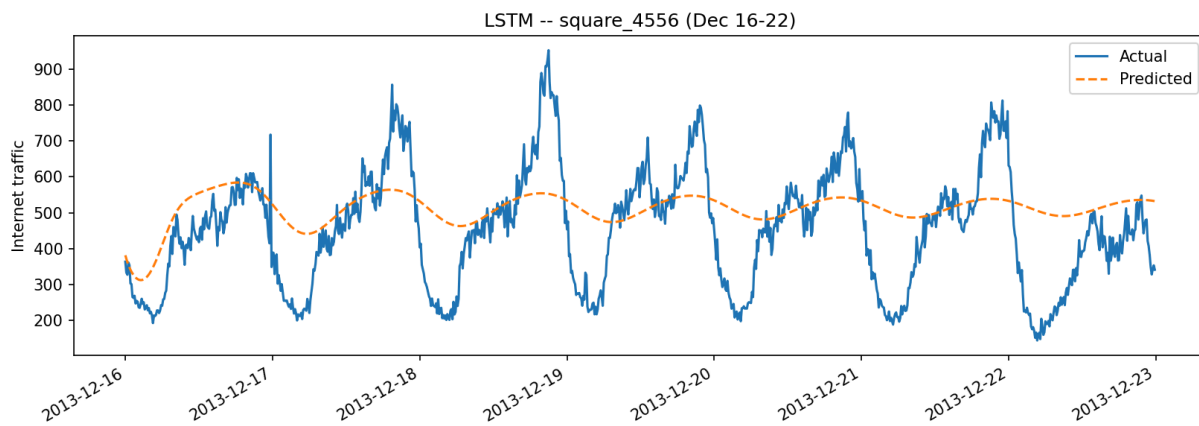


Figure 8c. LSTM forecast vs. actual, square 4556, Dec 16-22.

SARIMA again had the lowest MAE and RMSE: 112.64 and 131.88. GBM-fallback had the largest errors, while LSTM was between the other two models. GBM-fallback increasingly over-predicted later in the forecast.

6.4 Cross-area comparison

The results show that no single model behaved the same way in every area. SARIMA was strong and stable in the two lower-traffic areas. GBM-fallback performed particularly well in the busiest area but showed larger errors in the quieter areas. LSTM was especially weak in the highest-traffic area.

Average computational cost also differed greatly:

Model	Training time	Prediction time
SARIMA	6.666 s	0.028 s
GBM-fallback	1.342 s	7.597 s
LSTM	143.769 s	117.265 s

LSTM required much more computing time than the other two models. Its training time was about 21 times SARIMA's, while its prediction time was about 15 times SARIMA's.

7. Failure Analysis

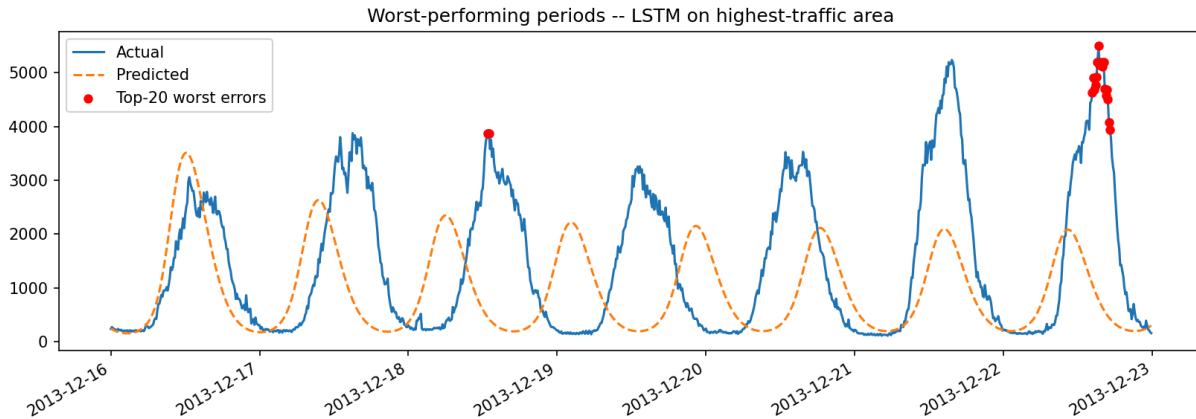


Figure 9. LSTM's 20 worst-error points, highest-traffic area, Dec 16-22.

The largest failure occurred when the LSTM forecast the highest-traffic area. Its biggest errors were concentrated around the final high peaks, where actual traffic reached about 5,000-5,500 units while the prediction stayed in a much smaller range.

One likely reason is the recursive forecasting setup. Each prediction becomes the input for the next prediction, so small early errors can build up. The LSTM also did not receive explicit time-of-day or day-of-week features. This means it had to learn the daily pattern from its own previous predictions.

A possible improvement would be to give the LSTM explicit cyclical time features, similar to those used by the GBM-fallback model. This was suggested for future work but was not tested in this study.

8. Conclusion and Future Work

This study compared SARIMA, gradient-boosted trees, and LSTM for Internet traffic forecasting across areas with different traffic characteristics.

The results show that the gradient-boosted model performed best in the busiest area, while SARIMA performed best in the two lower-traffic areas and was more consistent overall. LSTM performed poorly on the highest-traffic area and required much more computing time in this particular setup.

The results also show why model choice should depend on the traffic pattern of the area. A model that works well in one location may not work as well in another.

There are several limitations. The evaluation covered one week and only three areas. Forecasts were recursive over seven days, the hyperparameter searches were relatively small, and MAPE can be unstable when actual traffic is close to zero.

Future work should include explicit cyclical time features for the LSTM, direct multi-step forecasting, a wider hyperparameter search, evaluation on more weeks and areas, and external information such as public-holiday indicators.

References

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